ARX–RBF Motion Modeling for Cyberattack Detection and Recovery Monitoring in Industrial Robots

Muhammad Usman

COMSATS University Islamabad, Sahiwal Campus, Pakistan

Abstract

A robot can continue to deviate from its intended motion after an unauthorized command ends. A detector that follows recent measurements may lose evidence of that deviation, whereas an inaccurate protected reference may keep an alarm active after recovery. We study this trade-off using a command-driven motion reference, a learned residual correction and explicit alarm-clearance rules. Clean commissioning data select the reference and correction; normal calibration fixes thresholds before attack evaluation. Seventeen controls and ablations are evaluated on 189 synthetic episodes with model mismatch, legitimate command changes and unauthorized motion. The proposed nonlinear correction reduces benign alarm points from 7,156 to 7,012 relative to the uncorrected dual-window reference, but fails the declared improvement criterion: the reduction is 2.01%, one plant worsens, and pre-return gaps increase. The short window changes no alarm decisions. A following predictor achieves higher schedule F1 but loses evidence before physical return much more often. Five chronological validation folds show an inconsistent prediction benefit. Separate commissioning, uncertainty and recent author-code inertial-forecasting studies examine reference accuracy and recovery evidence. The results support distinguishing attack detection from physical return and reporting their costs together; they do not establish a novel detector or superiority over published robotic systems.

Keywords: digital twins; industrial robots; anomaly detection; recovery monitoring; system identification; model mismatch.

1 Introduction

An industrial robot does not necessarily return to its intended trajectory when an unauthorized command stops. Its position and velocity reflect the earlier disturbance, the controller's response and the plant dynamics. An alarm that clears as soon as prediction error falls can therefore disappear while the robot is still displaced. The opposite error is also possible: the robot returns, but an inaccurate reference keeps the alarm active. These two failures concern the meaning of the reference as much as the accuracy of the detector.

Digital twins provide a useful setting for studying this problem because they connect observations with a representation of the physical process. Recent surveys cover their roles in security monitoring and infrastructure lifecycle security [1, 2]. Those roles require more than a low forecasting error. A reference must represent the behavior relevant to the decision. For an attack alarm, that may be the motion the robot would have followed under approved commands, rather than the next motion most consistent with recent measurements.

Robotic anomaly detectors such as CASPER and its edge-computing successor learn regularities in external motion sensing [3, 4]. Hybrid robot twins also combine mechanical, control and learned components [5]. These approaches motivate prediction-based monitoring, but they do not make all prediction residuals interchangeable. An observation-following predictor may become accurate during a sustained unauthorized trajectory. A command-driven reference can retain the discrepancy, provided its clean initialization and approved-command history remain trustworthy. Its accuracy under legitimate changes then becomes central to false alarms and clearance.

We ask: *when does a command-driven motion reference provide useful evidence for clearing an alarm under model mismatch?* The study assumes a protected copy of approved commands and noisy but honest joint measurements. It tests an explicit candidate: a stable autoregressive reference with a frozen radial-basis correction, followed by short- and long-window return checks. The correction is fitted on clean commissioning data; it cannot learn from an attack during monitoring. This choice makes the information boundary inspectable, but does not establish either algorithmic novelty or physical safety.

The original investigation proceeds through five linked validation studies. An initial recovery experiment motivates a clean-data identification study; a separate inertial-sensing study checks predictor training under common inputs; a residual experiment isolates uncertainty in windowed return evidence; and a frozen candidate audit measures detection, false alarms, recovery and component effects together. Later designs use earlier outcomes as development evidence. They do not retroactively become independent confirmations of the earlier hypothesis. This distinction follows the broader need to separate model selection from security evaluation [6]. Subsequent chronological cross-validation and recent author-code forecasting runs extend normal-prediction assessment without replacing the frozen attack audit.

The comparisons jointly assess normal prediction, sustained attack evidence and clearance cost. Matched component removals distinguish reference accuracy from alarm persistence; a single F1 value cannot resolve these effects.

The contributions supported by the completed work are threefold. First, we specify and test a command-driven reference and alarm-state model with an explicit separation between observed residuals and evaluator-only physical return. Second, matched controls and ablations show that protected reference evolution, nonlinear correction and alarm persistence have different effects: the simpler reference retains the main benefit, while the proposed correction fails its improvement criterion. Third, commissioning and uncertainty diagnostics explain why better prediction alone may not yield usable return evidence. These are empirical and methodological findings. The present evidence does not validate a new state-of-the-art detector.

The following sections position the prior work, define the model and evaluation, and report the complete outcomes.

Derivations and supporting results remain in the article's appendices.

2 Literature Review

2.1 Robot monitoring and reference identification

CASPER couples external inertial sensing with context-aware industrial-arm anomaly detection [3]. The subsequent edge study develops real-time processing for related robotic monitoring [4]. Their sensing modality differs from the joint positions, velocities and protected commands used here. We therefore examine independently implemented CNN predictors on a common nominal inertial recording in a separate track. An error measured in that track cannot be interpreted as attack-detection superiority over either complete published system.

Ayankoso et al. combine robot mechanics, drives, control and learned uncertainty accommodation in a hybrid condition-monitoring twin [5]. Data-driven robot calibration is itself a broad research area [7]; adding a learned correction to a nominal reference is not sufficient evidence of novelty. Classens et al. address structured mechanical-system identification using instrumental variables and block-coordinate descent [8]. Our finite autoregressive search uses ordinary least-squares regularization and tests an application-specific recovery requirement, rather than extending that identification theory.

The voraus-AD study provides real robot executions and MVT-Flow, a normalizing-flow anomaly detector [9]. It is a relevant public comparator for an independent real-robot track because both data and author code can be inspected. Its published full-execution likelihood is an offline episode score. Comparing it directly with an online alarm-clearance delay would change the task. FactoryNet extends public industrial data coverage and exposes setpoints, effort, feedback and context in a common schema [10]. Its latest manuscript is a preprint, and its releases combine new acquisitions with repackaged sources. Independence must be assessed by acquisition, not by dataset name.

2.2 Forecasting, uncertainty and runtime decisions

ModernTCN examines convolutional time-series modeling [11], iTransformer attends across variates for forecasting [12], and TimeMixer uses decomposable multiscale mixing [13]. We execute their author model code on the common normal IMU prediction task. Their published forecasting benchmarks differ from robotic attack detection, so this transfer evaluates prediction under our declared inputs and budget. A more accurate predictor of corrupted motion can still be less useful for retaining attack evidence.

Interval-observer fault detection already treats model uncertainty and observer design explicitly [14]. FAIL-Detect calibrates uncertainty-aware runtime failure monitors from successful robot rollouts [15]. We include a constant episode-band control inspired by that calibration principle in the earlier experiment, without claiming to reproduce its policy-specific signal construction or full method. Howard et al. develop time-uniform confidence sequences [16]; our Gaussian window diagnostic uses simpler tail bounds and error spending under stated assumptions.

Kaisar et al. describe a robotic-vehicle cybersecurity architecture involving a kinematic twin, supervisory handover and consistency-gated recovery [17]. This is close conceptual prior art. The available abstract supports that overlap, but the full algorithm was inaccessible during our audit, leaving an equation-level comparison unresolved. RESCUE uses counterfactual interventions to evaluate the usefulness of runtime alarms [18]. Our counterfactual serves a narrower purpose: measuring when a simulated plant returns to the trajectory it would have followed without command corruption.

Evaluation also matters. TSB-AD examines the reliability of time-series anomaly-detection comparisons [19], while VUS studies accuracy measures for temporal anomalies [20]. We retain unadjusted point metrics and separately report event detection and recovery. We do not adopt point adjustment or optimize a threshold on test attacks.

2.3 Limitations and unresolved comparison requirements

The literature supports the individual ingredients of prediction, correction, uncertainty calibration and persistent alarm state. It does not justify treating their combination here as automatically new. The principal comparison gaps are specific: different sensing modalities for CASPER, a full-execution objective for MVT-Flow, unresolved implementation correspondence for some published systems, and missing physical-return labels in public fault data. These are limits on what our experiment can establish, not evidence that the cited systems are generally defective. A claim of superiority requires a compatible implementation and matched information access.

3 Methods

3.1 State, information and threat model

Let $x_t = (q_1, q_2, \dot{q}_1, \dot{q}_2)$ be the physical state and x_t^0 the state of the same plant under the complete approved command history without corruption. The monitor observes $y_t = x_t + n_t$ and approved position/velocity commands u_t. Its maintained state comprises predicted-state and command histories, residual buffers, an alarm bit and a return counter. The evaluator alone receives x_t, x_t^0 and the corruption schedule. Figure 1 shows this information boundary.

The attacker changes commands delivered downstream of a protected command copy. Clean commissioning data, the first 100 initialization observations and that copy are trusted. Measurements contain noise, but adversarial sensor forgery is outside this experiment. Once initialized, the protected reference consumes no new measured state. This prevents corrupted motion from changing its predicted history or fitted parameters. It is conditional on the protected inputs, and does not address compromise of both commands and telemetry.

The reference estimates the response to approved commands, initialized from the clean prefix and synchronized through command timing. It is an empirical motion reference, without online parameter adaptation or validated industrial physical fidelity.

3.2 Stable command-driven reference

The base is a clean-data autoregressive model with exogenous inputs (ARX). In its fitted coordinates,

$$b_t = a + \sum_{k=1}^{p} A_k b_{t-k} + \sum_{k=1}^{p} B_k u_{t-k}, \tag{1}$$

$$r_t^0 = \mu + s \odot b_t. \tag{2}$$

Commands retain their recorded physical coordinates. Training supplies state location μ and positive scales s; constant or nonfinite state channels are rejected. After initialization, Equation 1 uses predicted states in its own history. The plant-specific orders are 8, 16 and 32 for Q1–Q3, and each selected companion matrix has spectral radius below one. Stability concerns this fitted linear recurrence; it does not establish bounded physical error under arbitrary commands or guarantee transfer to a different robot.

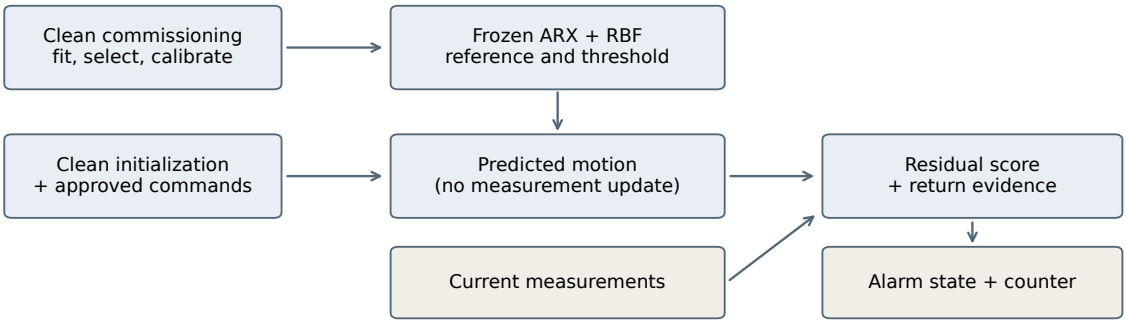


Figure 1: Tested reference and alarm model. Clean data determine the frozen recurrence, correction and threshold. Approved commands drive the reference after initialization. Measurements determine residuals and alarm state, without updating the reference. Counterfactual physical return is restricted to evaluation.

The following control instead uses measured history to predict the next observation. This changes the reference target while retaining the available channels.

3.3 Training-only residual correction

A correction estimates systematic normal residuals without feeding them back into the ARX recurrence. Its feature vector is

$$f_t = [r_t^0, u_{t-1}, u_{t-4}, u_{t-16}, u_{t-32}], \tag{3}$$
$$z_t = (f_t - \mu_f)/s_f, \tag{4}$$

with 20 dimensions. The feature means and scales use training observations only; a feature scale at or below 10^{-12} is replaced by one. This is distinct from rejecting a constant physical-state scale. Farthest-point selection in standardized training space deterministically chooses K centers c_i, starting from the observation closest to the origin. The bandwidth h equals a selected multiplier times the median positive center distance.

The radial basis functions (RBFs) and corrected output are

$$\phi_i(z) = \exp\{-\|z - c_i\|_2^2/(2h^2)\}, \tag{5}$$
$$r_t = r_t^0 + s \odot W^T[\phi_1(z_t), \ldots, \phi_K(z_t), 1]^T. \tag{6}$$

The four-output coefficient matrix minimizes

$$N^{-1}\|\Phi W - E\|_F^2 + \lambda\|W_{1:K,:}\|_F^2, \tag{7}$$

where E contains standardized clean residuals. The intercept is unpenalized, and nonfinite fits are rejected. A linear correction uses the same 20 standardized features and objective, providing a direct nonlinearity control. RBF heads contain $4(K+1)$ coefficients: 132 for 32 centers and 260 for 64 centers. These counts exclude stored centers, feature statistics and the base ARX. The linear head contains 84 coefficients.

Bounded activations and fixed weights bound correction magnitude, not physical prediction error. Weights, centers, scales and thresholds remain frozen during evaluation.

3.4 Entry score and alarm-clearance state

Define $e_t = (y_t - r_t)/s$ and $a_t = \max_j |e_{t,j}|$. Strict exceedance $a_t > \tau$ raises an alarm and resets its return counter. At other eligible points, the candidate checks

$$g_t = \mathrm{I}\{\|\bar{e}_t(5)\|_\infty \le 0.1 \ \wedge \ \|\bar{e}_t(41)\|_\infty \le 0.1\}, \tag{8}$$

where $\bar{e}_t(m) = m^{-1}\sum_{k=0}^{m-1} e_{t-k}$. Ten consecutive qualifying points clear an active alarm; a new entry exceedance has priority. The windows contain 0.10 and 0.82 seconds of samples at 50 Hz, and the dwell contains 0.20 seconds. Only complete windows after the initialization prefix are eligible. Signed means can cancel oscillatory deviations, so the gate measures agreement with the reference rather than certifying physical return.

Invalid history resets the evidence streak and cannot clear an active alarm; the implemented known-state mask then remains false. The final campaign admits complete finite episodes, so missing-data behavior is software-tested only. Algorithm 2 specifies the causal order.

3.5 Physical return and uncertainty

The evaluator measures $d_t = \max_j |x_{t,j} - x_{t,j}^0|/s_j$. Reference/noise bounds b_t, η_t would give $d_t \le a_t + b_t + \eta_t$. Sufficient return requires $a_t + b_t + \eta_t \le \delta$, which is infeasible when the allowance exceeds the tolerance. Empirical maxima under Gaussian noise are not deterministic bounds. Appendix A develops this limitation and the window diagnostic.

4 Experiments

4.1 Datasets and experimental populations

The study separates simulated attack/recovery evaluation, real-data normal prediction and admission of future public evidence (Table 3). They answer different questions and are not pooled into one performance ranking. Five linked studies document the development sequence (Table 5). Later choices use earlier outcomes as development; the earlier studies do not become independent confirmations of those choices.

4.1.1 Physical simulator and command construction

The plant is a horizontal two-joint point-mass arm:

$$M(q)\ddot{q} + C(q, \dot{q}) + B\dot{q} = \tau. \tag{9}$$

For $a = m_2 l_1 l_2$ and $d = m_2 l_2^2$, its inertia entries are $M_{11} = (m_1 + m_2)l_1^2 + d + 2a\cos q_2$, $M_{12} = M_{21} = d + a\cos q_2$, and $M_{22} = d$. The Coriolis vector is $a\sin q_2[-2\dot{q}_1\dot{q}_2 - \dot{q}_2^2, \dot{q}_1^2]^T$. A proportional–derivative controller requests

$$\tau^* = \mathrm{clip}\{K_p(q^c - q) + K_d(\dot{q}^c - \dot{q}), -20, 20\}, \tag{10}$$

with diagonal gains $(18, 12)$ and $(4, 3)$. With actuator lag, $\dot{\tau} = (\tau^* - \tau)/T_a$. Nominal masses are $(1, 0.8)$ kg, lengths $(0.5, 0.4)$ m and damping $(0.08, 0.06)$; Table 4 defines the actual plants. Position and velocity noise are independent zero-mean Gaussian with standard deviations 0.0005 rad and 0.002 rad/s before the plant multiplier.

Sampling is 50 Hz with eight fourth-order Runge–Kutta substeps per sample. A separate short integration check supports numerical consistency, not fidelity to an industrial robot. Gravity, contact, flexibility and thermal effects are absent. Approved positions are

$$q_1^c = a_1 \sin\theta + 0.025\sin(2.3\theta), \tag{11}$$
$$q_2^c = a_2 \sin(\theta + \phi) + 0.018\sin(1.7\theta), \tag{12}$$

Table 1: Algorithm 1: normal-only fitting, selection and freezing.

Input	Separate clean training, development and calibration episodes; declared candidate grid and tie-break rule.
1	Fit training-only state statistics and candidate ARX references; reject invalid or unstable candidates. Preserve each candidate outcome.
2	Generate free-running training predictions from clean initial histories. Construct the 20-dimensional features and fit their statistics and centers on training only.
3	Solve the ridge objectives. Evaluate each frozen candidate on the same normal-development episodes, keeping episode boundaries intact.
4	Within the linear and RBF roles separately, select by worst-episode residual $q_{.99}$, then mean RMSE, coefficient count and identifier. Preserve the chosen artifact without refitting on development.
5	Score the separate normal-calibration population. Set the fixed entry threshold to ascending rank 5,445 of 5,500 scores.
6	Seal models, transforms, thresholds, methods, seeds, inputs, code, budgets and criteria before generating the new evaluation conditions.

Table 2: Algorithm 2: prospective monitoring of one episode.

Input	Frozen model and threshold; clean 100-point prefix; approved commands and current measurements.
1	Initialize predicted history from the prefix; set alarm and evidence counter to zero.
2	Predict b_t from earlier predicted states and commands; compute r_t using the frozen correction.
3	Read y_t, calculate e_t, a_t, and update the complete residual windows. No label or future measurement is read.
4	If history is invalid, reset the streak and retain alarm state. Otherwise, an entry exceedance raises alarm and resets the streak.
5	If alarm is active without an exceedance, increment the streak when $g_t = 1$, otherwise reset it. Clear after the declared dwell.
6	Save score, validity and alarm at index t. Advance predicted history, leaving fitted parameters fixed. Continue across attack boundaries.

with analytic velocities under the causal time warp. The final tasks are N14: $\theta = 0.49t$, amplitudes $(0.28, 0.37)$, phase 0.93; O14: $\theta = 0.43t + 0.008t^2$, amplitudes $(0.33, 0.31)$, phase 0.72; and P14: $\theta = 0.94t + 0.18\sin(0.31t)$, amplitudes $(0.35, 0.38)$, phase 1.14.

The final audit crosses these three tasks, three existing plants, seeds 17, 42 and 2026, and seven conditions: ordinary operation, authorized offset, authorized slowdown, brief unauthorized offset, persistent offset, hold and slowdown. There are 81 wholly benign and 108 attack episodes. Each contains 1,200 samples, including the clean 100-point initialization. Corruption begins at index 347 and ends before 653, except the eight-point brief offset. Offset magnitude is 0.11 rad on joint 1; slowdown rate is 0.72. Authorized changes update the approved command, while attacks change only the delivered command. Truth and schedule remain outside the inference interface.

4.1.2 Real nominal recording and independent sources

The separate IMU study uses the provider-v2 nominal `IMU_10Hz.csv`, with accelerometer, gyroscope and magnetometer axes. It differs from the original CASPER-v3 archive. Exact correspondence to the published runs, timestamp units and acquisition-versus-arrival timing remain unresolved. We report sample-based prediction errors, not physical-time attack delays inferred from the filename.

voraus-AD metadata and original splits were verified, but its process signals remain unopened [9]. FactoryNet remains source-audited only [10]. Appendix C records hashes, permissions, source overlap and code-admission issues; neither source contributes measured model results here.

4.2 Preprocessing, chronology and calibration

4.2.1 Synthetic commissioning and held-out conditions

Three clean commissioning episodes per plant fit the base and corrections. Their rates are 0.35, 0.7 and 1.1, with amplitudes $(0.22+0.04i, 0.34-0.025i)$ and phase $0.5+0.3i$ for $i = 0, 1, 2$, covering unchanged, authorized-offset and authorized-slowdown motion. The earlier normal episode bank uses rate $0.32+0.047i$, amplitudes $(0.23+0.012(i \bmod 7), 0.29 + 0.013(i \bmod 9))$ and phase $0.4 + 0.06(i \bmod 11)$. Indices 0–8 select configurations and 9–13 calibrate. Five remaining archived checks are not new independent data.

Transforms, centers and regression coefficients use training only. Each episode retains its own initialization and history; windows never cross episode or split boundaries. After selection, each reference has 5,500 eligible calibration scores. The ascending score at rank $\lceil 0.99(5500) \rceil = 5445$ fixes its threshold. Strict exceedance handles ties. Calibration cannot choose models, seeds or a better percentile. Equal point-score quantiles do not equalize a stateful rule's alarm occupancy or guarantee future 1% false alarms.

The candidate check uses 81 new normal combinations based on tasks K13, L13 and M13. The attack audit freezes the fits, thresholds and 17-method roster before generating N14, O14 and P14. The tasks are new; the simulator and plants are reused. All resulting conditions are now exposed.

4.2.2 Chronological IMU partitions

Zero-indexed training rows $[219, 9185)$ precede development rows $[9185, 12007)$. Splitting occurs before the trailing width-20 median filter, training scaling and history creation. Each partition loses 19 filter warmup observations. Timestamps must increase strictly, with no gap exceeding five times the training median increment. Constant or nonfinite training channels are rejected.

All predictors share 8,192 training and 2,048 development targets after 755 history observations. Development targets are raw rows 9959–12006 inclusive. A later 755-residual aggregation has 1,294 aligned endpoints. The compact CNN uses only its 45-point history, but receives the same eligible target rows. These data were used for checkpoint selection and cannot serve as independent final evaluation.

4.3 Experimental setup and optimization procedure

4.3.1 Finite model search

The earlier reference search comprises ARX orders 4, 8, 16 and 32 and finite impulse response lengths 16, 32 and 64, crossed with ridge penalties $10^{-7}, 10^{-5}, 10^{-3}$ on each

Table 3: Data and information boundaries. Counts identify the populations actually used or admitted. New public-data receipt is not an additional experiment.

Source	Population	Permitted model information	Status and interpretation
Two-joint simulation, E5	3 training, 9 development and 5 calibration episodes per plant; 81 fresh normal and 189 audit episodes	Noisy joint position/velocity and protected approved commands; clean initialization	Synthetic, with evaluator-only command intervals and same-plant physical-return truth.
CASPER provider-v2 nominal IMU, E3	8,192 common training and 2,048 development targets; nine channels	Past causally filtered accelerometer, gyroscope and magnetometer observations	Real nominal prediction; no attack evaluation, unresolved paper-release and clock correspondence.
voraus-AD original 100 Hz release	2,122 executions; original split 948 training / 1,174 test	No process signals admitted to modeling yet	Metadata verified only; normal/fault labels, no established cyber or physical-return labels.
FactoryNet	Author-reported mixed real/synthetic corpus	No observations downloaded	Partially admitted source; imported sources are not new independent acquisitions.

Table 4: Fixed simulated plants. Mass and damping factors multiply the nominal values; noise multiplies both sensor standard deviations.

Plant	Mass factors	Damping factors	Lengths (m)	Lag (s)	Noise
Q1	(1.10,0.86)	(0.80,1.20)	(0.48,0.42)	0	1.0
Q2	(0.88,1.15)	(1.30,0.85)	(0.53,0.37)	0.03	1.0
Q3	(1.16,1.06)	(0.90,1.40)	(0.46,0.43)	0.06	1.5

plant: 63 fits. Stable ARX orders 8, 16 and 32 with ridge 10^{-7} are inherited by the correction study.

The correction grid has eight RBF settings per plant: $K \in \{32, 64\}$, width multiplier in $\{0.5, 1\}$ and ridge in $\{0.001, 0.01\}$. Two linear penalties use the same feature inputs, giving 30 fits overall. Within each correction role, selection minimizes the worst episode's 99th-percentile standardized residual, then mean episode RMSE, head coefficient count and identifier. All candidates and failures remain recorded. "Optimal parameters" here means selected within this finite normal-development procedure, not a global optimum or a choice based on final attacks.

4.3.2 Neural training and stopping

The independent CASPER-style CNN has four valid one-dimensional convolutions, each with 32 filters, kernel five and ReLU. Width-three pooling follows each pair; flattening and a nine-output dense layer give 39,977 parameters for 755-point history. The Edge-style CNN has two valid convolutions with eight filters and kernel three, width-three pooling, and a nine-output dense layer: 1,369 parameters for 45-point history. Both use zero-initialized output heads, disclosed as an implementation choice.

Adamax uses learning rate 10^{-3} and batches of 64 in chronological order without shuffling. A normal-development improvement greater than 10^{-6} selects a checkpoint; patience is five epochs and the cap is 100. Seeds 17, 42 and 2026 start afresh without optimizer transfer. Per-fit CPU admission is 240 seconds, with total limits of 1,200 seconds wall and 900 seconds CPU.

4.3.3 Five-fold chronological validation of ARX–RBF

After the original audit, we perform a retrospective normal-prediction check on the nine existing commissioning episodes. Each has 1,200 observations and is divided into six consecutive 200-point blocks. In fold $k \in \{1, \ldots, 5\}$, training uses $[0, 200k)$ in each episode. The following block supplies 100 known-normal observations for state initialization and 100 scored targets, at $[200k + 100, 200(k + 1))$. Windows never cross a split or episode boundary. Each plant combines the training prefixes of its three episodes; all three plants therefore contribute 900 scored observations per fold.

Both the ARX control and ARX–RBF refit their means, scales, coefficients and correction features using only that fold's training prefixes. Orders, centers, widths and ridge penalties retain the earlier settings. Validation measurements cannot change fitted parameters. Unstable or nonfinite fits would remain recorded as failures with zero coverage. No thresholds or attack metrics are computed. Because these settings and commissioning data were used previously, this is a developmental stability check, not nested cross-validation or independent confirmation. Expanding training prefixes overlap, and each validation trajectory has its own 100-point initialization.

4.3.4 Recent author-code forecasting runs

ModernTCN, iTransformer and TimeMixer use their original model classes with 96 past IMU observations and a one-step, nine-channel forecast. All receive the same 8,192 training and 2,048 development targets as E3, with its unchanged training-only scaler and causal preprocessing. Model-defined normalization uses the observed input window only; no future covariates or current target enter the predictor. The original CNNs retain their disclosed 755- and 45-point histories.

Transferred settings follow the authors' ETTh1 examples: ModernTCN has one 64-wide block, kernels 51/5, patch size eight, stride four and dropout 0.3; iTransformer has two 256-wide layers, eight heads and feed-forward width 256; TimeMixer has two layers, width 16, feed-forward width 32 and three average-downsampling levels. Adam rates are 10^{-4}, 10^{-4} and 10^{-2} respectively. Batches of 64 follow chronological order. Each of seeds 17, 42 and 2026 starts afresh; normal-development MSE selects checkpoints with tolerance 10^{-6}, patience five and a ceiling of 100 epochs. Each seed has a 150-second CPU admission budget, reserving the estimated cost of the next epoch. Actual stopping and coverage are reported, rather than treating a budget-limited fit as converged. Compatibility pilots allow at most two epochs and do not select hyperparameters.

4.3.5 Execution environment and validation sequence

Runs use macOS/arm64, Python 3.12.15 and NumPy 1.26.4; the independently implemented CNNs use TensorFlow 2.16.2. The recent author-code study uses PyTorch 2.5.1 on an Apple M5 Pro with 24 GB RAM, one Torch CPU thread and deterministic algorithms. No GPU benchmark is reported. Inference outputs are sealed before joint label evaluation. Exact execution records and failures are retained.

5 Evaluation Metrics

5.1 Prediction and point-level detection

For N eligible targets and d training-standardized channels, normal prediction error is

$$\mathrm{MSE} = \frac{1}{Nd}\sum_{t,j}\left(\frac{y_{t,j}-r_{t,j}}{s_j}\right)^2, \qquad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}. \tag{13}$$

Development also uses the 99th percentile of $a_t = \|e_t\|_\infty$. A low normal error is not an attack-detection result. The same training scales are retained for each model's assigned evaluation.

Point counts compare the alarm bit with the command-corruption schedule on all eligible observations, without point adjustment:

$$P = \frac{TP}{TP+FP}, \quad R = \frac{TP}{TP+FN}, \tag{14}$$

$$F_1 = \frac{2TP}{2TP+FP+FN}. \tag{15}$$

Precision is zero when there are no positive predictions. Average precision (AP) uses continuous entry scores: $\mathrm{AP} = \sum_k (R_k - R_{k-1})P_k$, grouping equal scores at common thresholds. It is undefined with no positive labels. Gate variants can therefore share AP while producing different alarm trajectories. The final audit has 207,900 eligible points and 25,002 schedule-positive points per method, with complete score coverage.

5.2 Benign alarms, event coverage and delay

Benign occupancy is $N_{\mathrm{alarm}}/N_{\mathrm{benign}}$ on wholly benign episodes. Starts per hour divide alarm rising edges by $N_{\mathrm{benign}}\Delta t/3600$, where $\Delta t = 0.02$ seconds. The final benign population is 89,100 points, or 0.495 eligible hours. Short repeated alarms and one sustained alarm may have similar occupancy but very different start rates.

An overlap hit is any alarm during an attack interval. A fresh hit requires a rising edge within that interval. Both denominators are all 108 events; a preexisting alarm can overlap without a fresh onset. Delays measure samples from schedule onset to the first qualifying overlap or rising edge, multiplied by Δt. Misses remain explicit, with undefined delays rather than silently removed or assigned zero. There is one scheduled intervention per attack episode; these are paired simulation conditions, not independent industrial incidents.

5.3 Physical return and clearance

Physical return is the first start of ten consecutive post-corruption states with $d_t \le 0.1$. This retrospective evaluator label uses the uncorrupted same-plant trajectory and never enters inference. Timely clearance is an inactive alarm at or within 250 samples after that return. The ratio conditions on detected events for which physical return exists; an already inactive alarm has zero clearance delay. This definition must be read alongside early clearance and misses.

The registered early-clear metric records a falling edge after first overlap but before stable physical return. Gap points count inactive observations between that hit and return, or the episode end when no return exists. A continuity break near onset can count even while instantaneous deviation is small. An explicitly post-study diagnostic separately checks true deviation at early falling edges; it does not replace the registered outcome.

5.4 Predeclared improvement and cost

RBF dual must preserve or improve hits, fresh hits and timely clears on each plant, while not increasing early-clear events, benign alarm points or pre-return gaps. It must also reduce pooled benign points or gaps by at least 10% relative to base dual. These are descriptive registered requirements, not statistical noninferiority tests. Secondary metrics cannot rescue a failed primary criterion.

Wall time, worker CPU, sampled resident memory and output size measure whole recorded runs. Shared computation prevents interpreting total time divided by the number of methods as per-method latency. Stochastic seeds describe computational/noise variation within the same experimental systems; no confidence interval is presented as if they were independent robot acquisitions.

6 Comparison Methods and Baselines

6.1 Protected references and component removals

The final roster compares base ARX, linear correction and RBF correction under matched inputs. Point rules directly threshold a_t. Return rules retain alarm using a one-point residual condition, the 41-point window, or the 5/41 dual window, generally with ten-point dwell. The no-dwell variant requires one qualifying point. Base long versus base dual and RBF long versus RBF dual isolate the short window; RBF dual versus its no-dwell variant isolates persistence. Base dual versus RBF dual is the primary correction comparison. Every variant keeps its declared entry score, threshold and support.

The RBF-following ablation applies the correction trained on free-running features to measured-history predictions without refitting. It tests that disclosed assembly, not an optimally fitted following model. Table 11 retains all 17 methods.

6.2 Ordinary prediction and accumulation controls

Persistence predicts the last measured state. The causal rolling mean predicts from the preceding 21 observations, excluding the current target. A one-sided CUSUM uses $G_t = \max(0, G_{t-1} + a_t - k)$, with k equal to the training-score mean plus its standard deviation. Its calibrated statistic has no reset on alarm in this registered implementation. Its persistence is therefore a deliberate comparator behavior, not a claim that all CUSUM schemes cannot recover. All controls use identical eligible episode support and cannot access labels or truth.

The IMU controls are training mean, persistence and AR(4), alongside the CNNs specified in Section 4. AR fits standardized channels with ridge 10^{-3} on mean normal equations, leaving its intercept unpenalized. Deterministic controls run once and neural models use all three seeds.

6.3 Recent published methods and correspondence

The recent-paper comparison window is 16 October 2023–16 October 2026, with preference for 2024–2026. Table 6 separates executed architectural transfers from an unexecuted candidate. Publication recency does not establish comparability or a current state-of-the-art ranking. Classical AR, persistence, rolling means and CUSUM remain explanatory controls. The two older mathematical references support observer and uncertainty background; they are not recent detector baselines.

Table 5: Five original studies and their comparison roles. They are a development sequence, not five independent robot datasets.

ID	Question	Evidence	Declared outcome
E1	Recovery under mismatch	216 synthetic episodes; 13 rules	Uncertainty allowance reduces alarms but prevents many clears.
E2	Clean reference accuracy	63 fits; 81 fresh normal episodes	Better fitting improves allowances; overall feasibility still fails.
E3	Common-input prediction	Nominal CASPER IMU prefix; nine outcomes	AR(4) has lowest development MSE; compact CNN reaches its epoch cap.
E4	Window uncertainty	36 synthetic residual traces; six rules	Required normal feasibility holds on 20/36 traces.
E5	Candidate and components	30 fits; 81 normal and 189 audit episodes	Nonlinear dual-window candidate fails its declared advantage criterion.

Table 6: Recent published methods: dates, actual execution and scope. No published headline score is substituted for a measured result.

Method	Publication	Execution here	Interpretation
CASPER [3]	ACM TIOT, 2024	Independent CNN implementation	Normal IMU prediction; complete attack detector not reproduced.
Edge CNN [4]	IEEE IoT J., 2025	Independent compact CNN	Same IMU targets; bounded training and modified protocol.
ModernTCN [11]	ICLR, 2024	Author model code	Convolutional forecast transferred to normal IMU prediction.
iTransformer [12]	ICLR, 2024	Author model code	Inverted-attention forecast on the same normal targets.
TimeMixer [13]	ICLR, 2024	Author model code	Multiscale mixing on the same normal targets.
MVT-Flow [9]	IEEE T-RO, 2024; preprint Nov. 2023	Source/data audit only	Whole-execution fault score; separate offline candidate, no measured ranking.

CASPER and its successor motivate the CNN architectures, but E3 uses independent implementations, a different normal-only protocol and a provider release whose original-run correspondence is unresolved. It is not a reproduction of their complete attack detectors. Their external IMU inputs also differ from the joint telemetry and protected commands in E5. Published headline metrics are not inserted into our measured ranking.

The three recent forecasters execute unchanged author model sources under our past-only prediction adapter, preprocessing, horizon and training budget. They test a different question from the protected command-driven reference and are not pooled into its attack/recovery ranking. MVT-Flow remains a candidate for a separate offline real-robot study: full-execution likelihood, symmetric temporal convolutions and unresolved objective/layer correspondence prevent immediate admission as an online recovery comparator. A compatible complete published-detector reproduction remains unfinished.

7 Results

7.1 Parameter selection and normal prediction

The 63-fit E2 search selects ARX orders 8, 16 and 32 with ridge 10^{-7}. Its complete normal comparison is retained in Appendix B. For the E5 correction, all 30 candidates produce eligible outcomes. Figure 2 displays every development criterion, including weaker settings. Selection chooses 32 centers on all plants, width multipliers 1, 0.5 and 1, and penalties 0.01, 0.001 and 0.001 for Q1–Q3. All selected linear penalties are 0.01. These are finite normal-development choices, not parameter choices based on the attack audit.

On 81 fresh normal episodes, RBF correction changes RMSE by +0.0050%, −1.2205% and −0.1002% for Q1–Q3. Corresponding tail changes are −0.1314%, −1.4022% and +0.1506%. The all-plant improvement requirement fails. Better performance on selected development episodes does not imply improvement on every new condition, and normal return-evidence availability is not post-attack clearance.

E3 tests learned prediction separately (Table 8, Figure 3). AR(4) achieves MSE 0.006074. All CASPER-style seeds stop at epoch 27 and select epoch 22, with MSE 0.017451–0.017703, 46.11–58.12% below their shorter pilot errors. Compact CNN errors are 0.042700–0.044730 and still improve at epoch 100.

Median filtering changes the prediction task and induces dependence. These normal-development errors support retaining ordinary predictors, but cannot establish attack-detection superiority or independent generalization.

7.2 Recent author-code prediction results

All nine recent-model runs complete on the same 2,048 normal-development targets (Table 9). ModernTCN gives MSE 0.012207–0.013543; iTransformer gives MSE 0.020031–0.023680; TimeMixer gives MSE 0.011629–0.015218. All nine runs stop at their declared CPU admission limit, before the patience criterion. These values describe transferred configurations under a finite budget; they do not estimate each architecture's converged performance. Normal-development checkpoint selection also makes them unsuitable as independent test estimates. The models have 173,861, 817,153 and 58,033 parameters respectively. AR(4) remains a useful low-cost reference on these filtered targets; different histories and training budgets limit a broader architectural ranking.

7.3 Five-fold temporal validation

All 15 plant–fold fits complete, giving 4,500 scored observations per method with no failures. Table 10 reports every fold, pooling equal-sized plant populations after each fold's training-only standardization. The correction improves pooled error in folds four and five and in 5/15 plant–fold pairs. Its benefit is small and inconsistent. The first

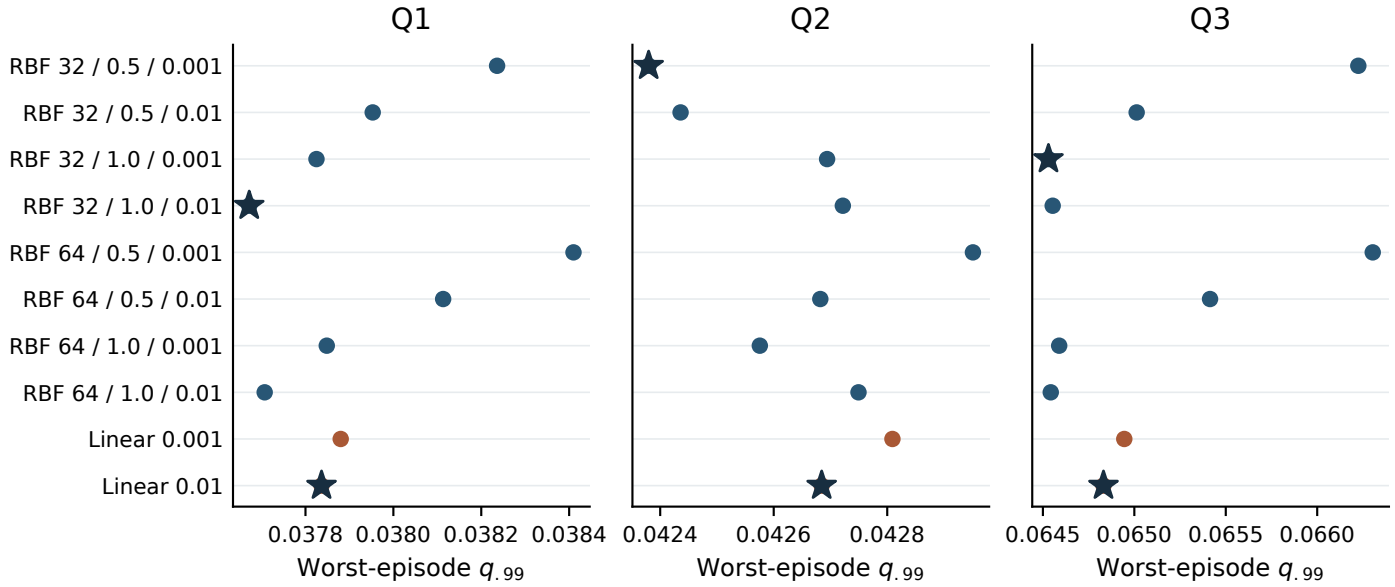


Figure 2: All 30 correction candidates on normal development: worst-episode 99th-percentile standardized residual (lower is better). RBF labels give centers/width multiplier/ridge; linear labels give ridge. Stars mark the selected candidate within each correction role on each plant. Horizontal scales differ across panels. These are selection outcomes, not held-out attack results.

Table 7: Selected E5 candidate and fresh-normal changes against the base reference. Positive error change is worse. Each selected RBF head has 132 coefficients, in addition to its stored centers and base model.

Plant	ARX p	Centers	Width	Ridge	τ	ΔRMSE (%)	$\Delta q_{.99}$ (%)
Q1	8	32	1.0	0.01	0.036966	+0.0050	-0.1314
Q2	16	32	0.5	0.001	0.037122	-1.2205	-1.4022
Q3	32	32	1.0	0.001	0.057400	-0.1002	+0.1506

Table 8: E3 normal-development prediction on the common filtered IMU prefix. Epochs are completed/selected. Every seed and deterministic control is shown; no attack metric is computed.

Predictor	Seed	MSE	Epochs	Stopping
Training mean	–	1.073987	–	Closed form
Persistence	–	0.019432	–	Closed form
AR(4)	–	0.006074	–	Closed form
CASPER-style CNN	17	0.017703	27/22	Patience
CASPER-style CNN	42	0.017451	27/22	Patience
CASPER-style CNN	2026	0.017458	27/22	Patience
Edge-style CNN	17	0.044730	100/100	Epoch limit
Edge-style CNN	42	0.043451	100/100	Epoch limit
Edge-style CNN	2026	0.042700	100/100	Epoch limit

Table 9: Recent author-code forecasts: all seeds, completed/selected epochs, standardized development MSE and per-seed fitting/scoring CPU. Stops: C, CPU admission; P, patience; E, epoch ceiling. No attack score is measured.

Model	Seed	Ep.	MSE	CPU (s)	Stop
ModernTCN	17	4/4	0.013426	129.9	C
ModernTCN	42	4/4	0.012207	130.3	C
ModernTCN	2026	4/4	0.013543	130.4	C
iTransformer	17	5/4	0.023680	135.1	C
iTransformer	42	5/5	0.020875	135.2	C
iTransformer	2026	5/5	0.020031	135.1	C
TimeMixer	17	8/8	0.014041	130.0	C
TimeMixer	42	8/7	0.015218	129.9	C
TimeMixer	2026	8/7	0.011629	141.7	C

fold uses only 200 training observations per episode; trajectory segment and fitted scale also change across folds, so the reduction in later-fold error cannot be attributed to training size alone. This diagnostic leaves the original fit selection and attack-audit outcome unchanged.

Table 10: Five-fold chronological normal validation. Lower standardized RMSE is better. Every fold uses 900 targets; the folds reuse overlapping training prefixes.

Fold	Targets	ARX RMSE	ARX–RBF RMSE
1	900	0.224181	0.224551
2	900	0.011501	0.011521
3	900	0.012052	0.012075
4	900	0.011103	0.011086
5	900	0.010940	0.010937

7.4 Detection and false-alarm trade-offs

Table 11 reports the complete frozen audit. RBF dual reduces benign alarm points from 7,156 to 7,012, a 2.0123% reduction below the required 10%. Q1 deteriorates from 2,397 to 2,522 points, and pooled pre-return gaps increase from two to three. The primary criterion therefore fails.

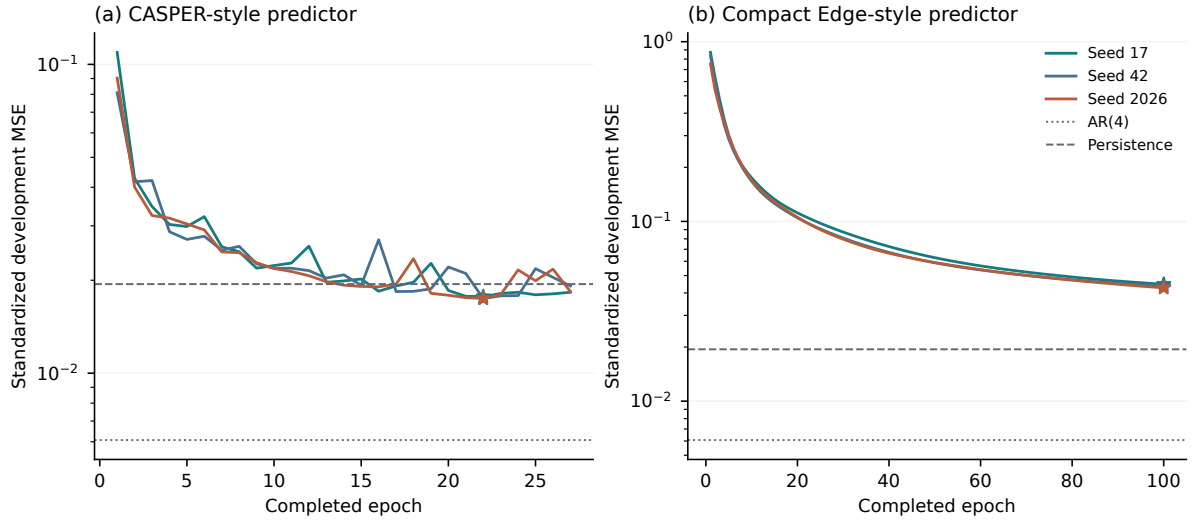


Figure 3: Saved E3 normal-development learning curves for all E3 neural seeds, with controls on the same targets. Patience stopping supports the adequacy of this bounded CASPER-style fit; the compact model remains limited by the epoch ceiling. These curves do not establish full-paper reproduction or attack-detection superiority.

Base and RBF dual overlap all 108 attacks; fresh hits increase from 101 to 105. This secondary gain cannot reverse the declared failure.

Figure 4 places schedule F1, benign occupancy and lost pre-return support beside one another. Following point has F1 0.9202 and AP 0.9501, exceeding RBF dual's 0.6407 and 0.7276. Yet it has an early falling edge on every attack, with 6,692 pre-return gap points. Following dual reduces these to 90 events and 2,970 points; base dual has one registered early-clear event and two gap points. CUSUM retains support but has no timely clearance. No single plotted axis establishes overall superiority.

7.5 Physical return and event interpretation

Figure 5 illustrates the distinction: schedule labels end with command corruption, while displaced motion can continue. Retaining alarm then supports recovery monitoring while reducing schedule precision.

The single RBF early-clear event occurs two samples after onset, with true standardized deviation 0.01649 below tolerance 0.1. A fresh alarm starts three samples later. It is a continuity break, not an outside-tolerance clearance. A separately labelled post-study falling-edge diagnostic finds no outside-tolerance early falls for protected-reference variants, versus 108 events for following point and 82 for following dual. This clarification preserves the registered metric and failed primary decision. Persistence misses 41 events and the rolling mean misses 60; their conditional clearance fractions cannot be presented as recovery success on every event.

7.6 Ablation and uncertainty studies

Removing the short window changes zero alarm entries in all 189 episodes. Its contribution is unsupported under these conditions. Removing dwell reduces RBF benign points from 7,012 to 1,195, with 108 overlap hits and 108 timely clears. Its registered early-clear count remains one, on a different episode. Figure 6 separates the component effects. The favorable no-dwell result is an exposed secondary finding; promoting it as a new confirmed detector would require a separately frozen study.

E1 reduces benign points from 621 to 309 using an empirical allowance, but clears only 36/108 detected events with return; its same-entry no-allowance control has 140 benign points and 108 clears. E2 reduces the harder plants' allowances by 51.27% and 38.56%, yet fails overall feasibility. Appendix B retains every earlier comparison under its distinct calibration.

E4 certifies enough normal endpoints on 20/36 constructed traces, failing its all-traces criterion. Averaging competes with temporal correlation and within-window motion. Appendix A gives the complete settings, derivation and outcomes; these are direct synthetic residuals, not attack trials.

7.7 Computational cost and verification

E5 normal development takes 16.115 seconds wall and 15.189 seconds worker CPU, with 53.23 MiB sampled peak memory. The final audit takes 27.515 seconds wall and 26.656 seconds CPU, with 90.17 MiB sampled memory and 83.44 MiB outputs. The IMU study takes 764.794 seconds wall, 753.252 seconds CPU and 553.05 MiB sampled memory. These are whole-pipeline measurements with shared work, not isolated inference latency or real-time guarantees.

The recent forecast campaign consumes 1208.167 seconds wall and 1199.209 seconds summed worker CPU, with 521.44 MiB maximum sampled memory and 14.48 MiB monitored outputs. All nine saved checkpoints reproduce their predictions exactly; 294 artifact and replay checks pass. The separate five-fold diagnostic uses 0.485 seconds wall and 0.297 seconds CPU and passes 109 saved-artifact checks. These are pipeline costs, not isolated inference latency.

The final audit passes 25 synthetic software tests and 30,448 saved-artifact checks, including inference replay and separate metric arithmetic. All 3,213 method–episode outcomes remain recorded. These checks support consistency with the declared implementation; they are not independent scientific replication. Appendix C records budgets, earlier costs and preserved failures.

8 Discussion

8.1 What the comparisons establish

The central distinction is between following measured behavior and preserving a reference to approved behavior. Protected evolution retains deviations in this simulator, but the ordinary ARX already provides that benefit. The nonlinear correction and dual-window assembly do not meet the registered improvement target. Existing identification, residual correction, averaging and hysteresis remain close prior art; combining them does not by itself establish an original detector.

Reference accuracy, entry scoring and clearance logic have different roles. The no-dwell contrast shows that

Table 11: All 17 registered E5 methods on 189 synthetic episodes. Benign uses 89,100 points. Hit/fresh means overlap/fresh-onset events out of 108; early is the registered continuity-based count. Clear is timely clears/detected events with return. AP evaluates continuous entry scores, not retained alarm state.

Method	F1	AP	Benign	Hit/fresh	Early	Gaps	Clear
base point	0.7809	0.7276	853	108/108	14	38	108/108
base return	0.6452	0.7276	6,811	108/101	1	2	108/108
base long	0.6381	0.7276	7,156	108/101	1	2	108/108
base dual	0.6381	0.7276	7,156	108/101	1	2	108/108
linear point	0.7799	0.7276	895	108/108	14	38	108/108
linear dual	0.6336	0.7276	7,498	108/101	1	2	108/108
RBF point	0.7816	0.7276	835	108/108	14	39	108/108
RBF return	0.6482	0.7276	6,652	108/105	1	3	108/108
RBF long	0.6407	0.7276	7,012	108/105	1	3	108/108
RBF dual	0.6407	0.7276	7,012	108/105	1	3	108/108
RBF dual, no dwell	0.7644	0.7276	1,195	108/108	1	1	108/108
following point	0.9202	0.9501	867	108/108	108	6,692	108/108
following dual	0.7070	0.9501	7,635	108/105	90	2,970	108/108
RBF following dual	0.7069	0.9501	7,678	108/105	90	2,979	108/108
persistence	0.0333	0.1016	527	67/67	67	15,742	67/67
mean (21)	0.0082	0.0867	84	48/48	48	11,545	48/48
base CUSUM	0.4198	0.1967	857	108/108	0	0	0/108

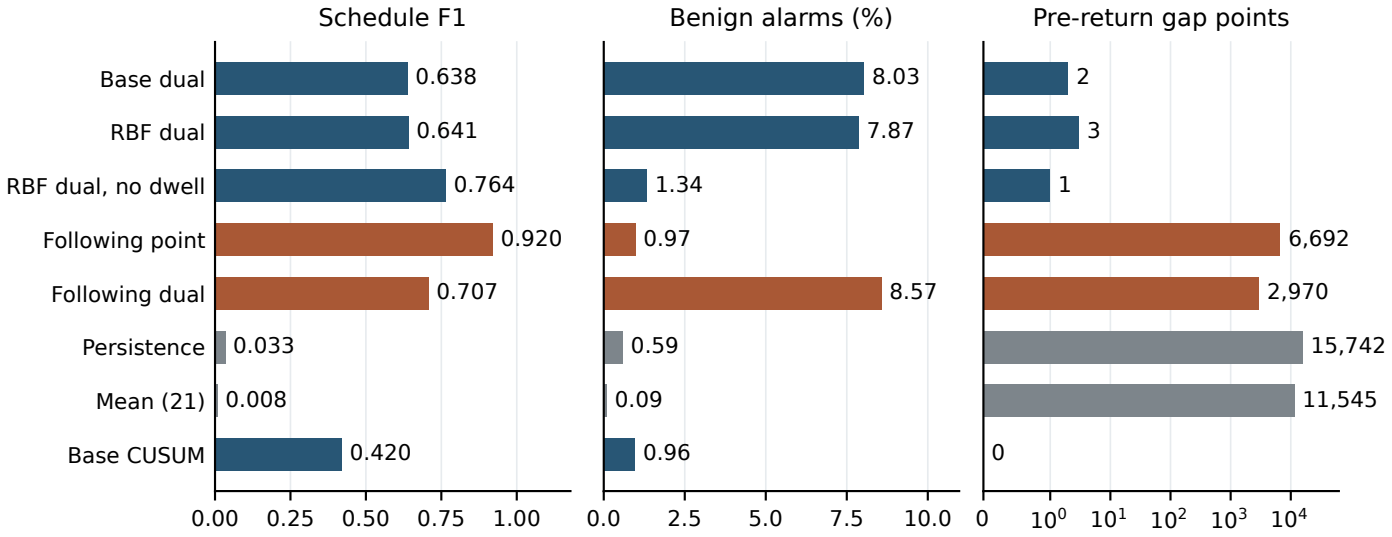


Figure 4: Selected contrasts from the complete E5 roster: corruption-schedule F1, benign alarm occupancy and lost alarm support before physical return. Bar labels give saved values; gap axes use a logarithmic scale with a linear region near zero. The figure is a descriptive presentation of the frozen outcomes, not a newly selected primary comparison.

persistence can dominate benign occupancy despite unchanged scores and thresholds. AP hides alarm retention; schedule F1 hides much of physical recovery. Both must be accompanied by event and clearance measures.

8.2 Relationship to published systems

CASPER and its successor motivate external-sensor prediction; existing hybrid twins and identification methods already establish modeling and correction. E3 does not reproduce their full attack detectors. The Gaussian argument is standard error spending, and the inaccessible full Kaisar et al. architecture leaves an equation-level prior-art comparison unresolved.

MVT-Flow provides a concrete public route to an independent real-robot comparison, but its offline likelihood target differs from online physical-return monitoring. A new comparison must establish compatible features, supervision and output units before claiming that one method improves another. A fault label or a setpoint column alone cannot supply missing attacker semantics or trusted-command provenance.

8.3 Limitations and next evidence

Three parameterizations share one simplified simulator. Noise seeds repeat stochastic realizations, not independent plants or acquisitions. Missing physics, trusted initialization and honest sensing limit external validity. The residual diagnostic receives known noise/bias/motion bounds; no operational procedure for estimating all of them is validated. Normal-only IMU development and metadata-only public-data admission do not provide a real-robot recovery test.

voraus-AD supports a separate offline track after features and partitions are frozen. FactoryNet requires source-overlap and release checks. Compatible published-method reproduction and a worthwhile distinct mechanism remain open. The exposed outcomes constrain the next design; they cannot also confirm an outcome-informed replacement.

9 Conclusion

This study examines how a command-driven motion reference can support cyberattack detection and recovery monitoring when the modeled robot differs from the simulated plant. Protected reference evolution preserves evi-

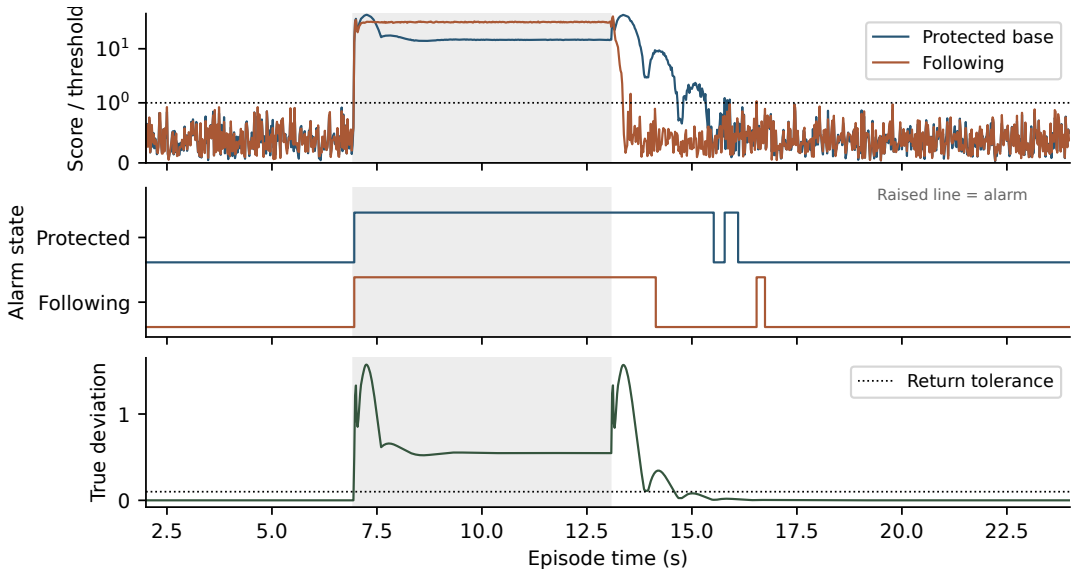


Figure 5: Post-study illustration from task N14, plant Q2, seed 17, persistent offset. Shading marks command corruption. The following residual falls while physical deviation remains above tolerance. This trace was selected to explain the aggregate contrast; it is not independent confirmatory evidence.

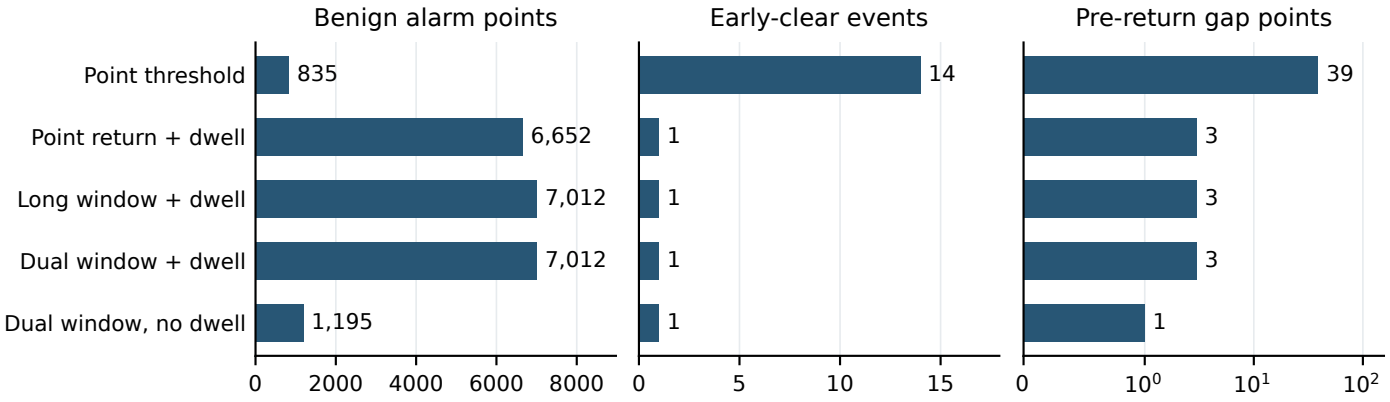


Figure 6: Matched RBF alarm-rule variants in E5. Entry score and calibration threshold are shared. The 41-point and dual windows produce identical alarm arrays. Dwell substantially changes benign occupancy; a small gap count alone does not account for missed events or late clearing. All variants detect and timely clear 108 events in this campaign.

dence that measurement-following prediction can absorb, while inaccurate references and persistent alarms impose a benign-operation cost. In the frozen attack audit, the nonlinear correction reduces benign alarm points by 2.01%, below the declared target, and the additional short window changes no decisions. Five chronological refitting folds also show an inconsistent prediction benefit: ARX–RBF improves pooled error in two folds and in five of the fifteen plant–fold comparisons.

The practical finding is that prediction error, attack coverage and alarm clearance must be assessed together. The study provides an explicit model, common-information controls, component tests and traceable evaluation of these competing objectives. Recent author-code forecasters extend the normal-prediction comparison; their training budget and separate sensing task remain part of its interpretation. The evidence does not establish a novel superior detector. Further work should test reference adequacy across independent recordings and evaluate compatible published systems under the same threat model before making broader deployment or generalization claims.

Declarations

Funding and competing interests. No funding or competing interests were declared.

Data and code availability. Versioned code, configurations, results and a reproducibility inventory are retained locally. No public repository release is claimed. Third-party data remain subject to provider terms; Appendix C records source identifiers and hashes. All supporting material discussed here is incorporated in this article.

AI assistance and author review. OpenAI Codex assisted with literature searches, code, experiment execution, analysis, writing and editing. The named author must review the scientific claims and approve the manuscript before submission. No completed human-contribution attestation or independent human review is asserted here.

A Uncertainty and feasibility

A.1 Conditional geometric bounds

Let $\|v\|_s = \max_j |v_j|/s_j$, $d_t = \|x_t - x_t^0\|_s$ and $c_t = \|y_t - r_t\|_s$. If $\|x_t^0 - r_t\|_s \le b_t$ and $\|n_t\|_s \le \eta_t$, the triangle and reverse triangle inequalities give

$$\max(0, c_t - b_t - \eta_t) \le d_t \le c_t + b_t + \eta_t. \tag{16}$$

Scalar errors aligned in the same direction attain these extremes. For coordinate error boxes with half-widths $b_{t,j}, \eta_{t,j}$, return within δ holds for every admissible pair

exactly when

$$\max_j \frac{|y_{t,j} - r_{t,j}| + b_{t,j} + \eta_{t,j}}{s_j} \le \delta. \tag{17}$$

Each coordinate can attain the summed magnitude, proving necessity as well as sufficiency. If any $b_{t,j} + \eta_{t,j} > \delta s_j$, the condition is infeasible. E1 instead tests $c_t + b \le \delta$ with a maximum observed normal discrepancy b. That maximum combines model error and noise; Gaussian noise is unbounded. The tested rule is therefore an empirical heuristic, not a verified certificate. Dwell cannot repair $b > \delta$.

A.2 Covariance-aware window evidence

For dimensionless residuals write $e_{t,j} = v_{t,j} + \beta_{t,j} + \epsilon_{t,j}$, with $|\beta_{t,j}| \le B_j$ and $|v_{t,j} - v_{t-1,j}| \le L_j$. Assume stationary zero-mean Gaussian AR(1) noise with marginal standard deviation σ_j and correlation $0 \le \rho_j < 1$. Its window-mean variance is

$$V_{m,j} = \frac{\sigma_j^2}{m^2}\left[m + 2\sum_{k=1}^{m-1}(m-k)\rho_j^k\right]. \tag{18}$$

For K registered windows and d channels, allocate $\alpha_t = 6\alpha/(\pi^2 t^2)$ on a clock that never restarts at alarms, and define

$$U_{t,j}(m) = |\bar{e}_{t,j}(m)| + B_j + \frac{L_j(m-1)}{2} + \sqrt{2V_{m,j}\log(2dK/\alpha_t)}. \tag{19}$$

With probability at least $1-\alpha$, $|v_{t,j}| \le U_{t,j}(m)$ for every channel, complete registered window and prospective time within a trace. Thus coordinatewise return is certified on that event if

$$\min_{m\in\mathcal{M}} U_{t,j}(m) \le \delta_j \quad \text{for every } j. \tag{20}$$

To verify the claim, the two-sided Gaussian tail is at most $\alpha_t/(dK)$. A union bound over channels, windows and times uses $\sum_t \alpha_t = \alpha$. Window overlap and cross-channel dependence do not invalidate that bound. Finally, $|v_{t,j} - \bar{v}_{t,j}(m)| \le L_j(m-1)/2$ and $|\bar{\beta}| \le B_j$ yield Equation 19. Different channels may select different windows because all bounds hold simultaneously.

This is elementary error spending under supplied assumptions, not new confidence-sequence theory. Invalid observations invalidate affected windows without resetting time. Failure to certify leaves return uncertain and does not generate an attack alarm. Unknown covariance, heavy tails or nonstationarity fall outside the guarantee.

A.3 Residual experiment and complete results

E4 fixes $d = 4$, windows $\{1, 41, 201\}$, $\alpha = 0.05$, tolerance 0.1 and bias bound 0.01. The four (σ, ρ) settings are $(0.03, 0)$, $(0.06, 0)$, $(0.03, 0.8)$ and $(0.06, 0.95)$, with channel multipliers $(1, 0.8, 1.1, 0.9)$. Bias is $0.01\cos(0.003t + \phi_j)$ for phases $(0, 0.4, 0.8, 1.2)$. Deviation is zero, $0.02\sin(0.004t + \phi_j)$, or $0.02\sin(0.04t + \phi_j)$; its supplied increment bound is $2a|\sin(\omega/2)|$.

Each trace has 1,200 points and stationary initial noise. Seeds 17, 42 and 2026 supply paired innovations across settings, producing 36 constructed traces. Common support is points 201–1,200. Three fixed-window controls and their simultaneous envelope receive identical residuals/bounds; two deliberately misspecified controls set $\rho = 0$ for windows 41 and 201. All 216 outcomes remain recorded. The envelope must certify at least 95% of common points on every trace. The per-trace probability statement is not simultaneous 95% coverage across the experiment matrix.

The envelope passes 20/36 traces. Instantaneous evidence certifies none. Under independent low noise with faster variation, window 41 certifies every point while 201 certifies none: the longer drift allowance outweighs averaging. Under correlated low noise with slow variation, fractions are 100%, 85.8% and 100%; their 95.27% average does not satisfy the all-traces rule. At $(0.06, 0.95)$ no valid control certifies even settled motion. The misspecified 201-point control would certify 91.53% on average there, without a valid covariance guarantee.

This generator constructs residuals directly and supplies the bounds; it does not simulate robot dynamics or attacks. All deviations are within tolerance, so the absence of observed bound violations cannot establish rare unsafe-clearing risk. Operational estimation of the assumptions and behavior near the tolerance boundary remain untested.

B Earlier comparisons and full final metrics

B.1 E1: original recovery comparison

E1 comprises 216 episodes and 13 methods, with 237,600 eligible points and 135 intervention events. Twenty-seven events have missing command context. Nineteen clean episodes provide 20,900 eligible calibration points; the principal state-machine threshold search allows at most 209 alarmed points on this finite population. This differs from the final audit's point-quantile calibration. The additional constant episode band uses rank 19 of 19 episode maxima; shifted tasks do not establish exchangeability or conformal coverage. Matched-entry ablations preserve their own realized alarm burdens.

B.2 E2: normal commissioning

The finite grid contains ARX orders 4, 8, 16 and 32 and finite impulse response lengths 16, 32 and 64, each with ridge 10^{-7}, 10^{-5} or 10^{-3}, on three plants: 63 fits. Fitting and selection precede 81 new normal episodes. The selected orders are 8, 16 and 32 with ridge 10^{-7}. Table 15 retains distinct prediction roles. Forced-latch probes measure normal clearance feasibility, not post-attack recovery. The hard-plant reductions in empirical allowance do not establish a probabilistic or deterministic error bound.

B.3 E5: complete additional metrics and plant strata

The following tables complement the complete 17-method audit in Table 11. Alarm starts per hour use the total eligible duration of wholly benign episodes. Their high frequency for point rules reflects repeated short excursions; the metric should be read alongside occupancy. No unknown-state denominator trimming is applied to this audit. All 189 episodes and 3,213 outcomes are retained.

C Reproducibility and source identities

C.1 Recorded execution and failures

Versioned records retain inputs, configurations, commands, dependencies, code revisions, logs and every outcome. E1–E5 correspond respectively to `recovery_v0.8`, `commissioning_v0.9`, `casper_convergence_v0.11`, `return_bound_v0.12`, and `residual_twin_v0.13` followed by `attack_audit_v0.14`. Their calibration policies are not interchangeable. The later five-fold diagnostic and author-code forecast study have separate frozen records; E1–E5 are unchanged.

E1 consumed 229.20 s wall, 228.07 s CPU and 167.25 MiB sampled memory; E2 consumed 14.75 s, 14.30 s and 113.97 MiB. E1's initial launch failed before initialization because monitoring was blocked. E2 stopped once on a reporting-key collision after selection; the recorded repair

Table 12: E4 simultaneous-envelope certificates on normal synthetic residuals. Ranges span three seeds with 1,000 common points each. Every trace must reach 95%; the overall criterion fails.

Noise setting	Deviation profile	Certificates (%)	Passes
Independent, low	Settled	100.0–100.0	3/3
Independent, low	Slow variation	100.0–100.0	3/3
Independent, low	Faster variation	100.0–100.0	3/3
Independent, high	Settled	100.0–100.0	3/3
Independent, high	Slow variation	100.0–100.0	3/3
Independent, high	Faster variation	17.2–25.6	0/3
Correlated, low	Settled	100.0–100.0	3/3
Correlated, low	Slow variation	85.8–100.0	2/3
Correlated, low	Faster variation	0.0–0.0	0/3
Correlated, high	Settled	0.0–0.0	0/3
Correlated, high	Slow variation	0.0–0.0	0/3
Correlated, high	Faster variation	0.0–0.0	0/3

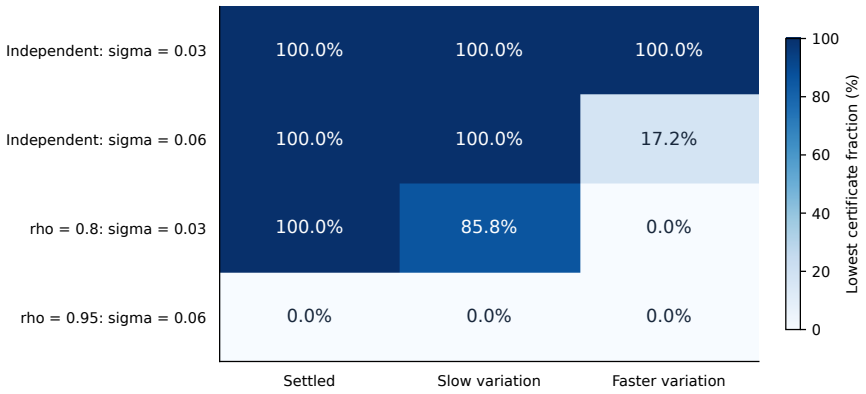


Figure 7: Lowest E4 certificate fraction across seeds in each residual condition. All true deviations lie within tolerance. Low certificate coverage means uncertain return under the supplied bounds, not detected attacks.

Table 13: E1 attack and recovery outcomes, retained from the saved presentation. Clearance counts condition on detected events with measured return.

Method	Benign	F1	Overlap/fresh	Early	Clear/return
Persistence	878	0.057	127/127	127	127/127
ARX one-step	792	0.803	108/108	108	108/108
Following point	794	0.808	108/108	108	108/108
Following hysteresis	993	0.785	108/108	108	108/108
Nominal hysteresis	789	0.713	108/108	45	108/108
Nominal fixed return	621	0.657	108/108	0	108/108
Learned point	819	0.700	108/108	18	108/108
Learned hysteresis	792	0.683	108/108	0	108/108
Learned fixed return	827	0.690	108/108	0	108/108
Uncertainty return	309	0.446	108/104	0	36/108
Episode CP band	0	0.744	108/108	75	108/108
Matched no allowance	140	0.709	108/108	0	108/108
Matched no dwell	20	0.453	108/108	0	36/108

preserved identical fits before fresh generation. E5 has no failed scientific attempt. Its final limits were 120 s wall, 90 s CPU, 1 GiB monitored memory and 128 MiB output. Worker peak memory was 99.53 MiB versus 90.17 MiB sampled; sampling is not an instantaneous hard bound. The IMU run used 2.68 MiB outputs and passed 17 synthetic tests and 3,754 saved-artifact checks. E4 used 0.565 s wall, 0.459 s CPU, 39.81 MiB memory and 11.07 MiB outputs, with 20 tests and 2,665 checks.

The final freeze SHA-256 is `b48a0c3927b465559a8f7b372c2d26ed7148642c97d2435de83995a0ad3be021`; its result-package manifest is `987d8904dda988c7ccd51d61a5e52cf8e162e8f0156286aad89c385813b785dc`. Asset receipts bind figures and tables to saved results. Identifiers establish provenance, not validity. Raw data and third-party manuscripts are excluded from the editable archive.

The added `fivefold_v0.16` diagnostic preserves its monitor-denied attempt and separately bound, scientifically unchanged repair. The `forecast_baselines_v0.17` study was approved for 1,800 seconds wall, 1,440 seconds CPU, 1 GiB sampled memory and 150 MiB output. All nine fresh fits stopped by CPU admission; no scientific retry or post-pilot configuration change occurred. ModernTCN, iTransformer and TimeMixer revisions are `56a9a2c018385`, `c2426e68ca13` and `e24610583b36`; licenses are MIT, MIT and Apache-2.0. Full identities, dependencies, hardware and commands remain in the execution records.

C.2 Real-data identity and comparator admission

The author-linked CASPER-1-and-2 provider-v2 nominal IMU file has SHA-256 `1221df9be80c229969f3f21f820837e68ad28f4ccb3949bf4013f00d6d69d62c`. Its original-paper run and clock correspondence remain unresolved.

Table 14: E1 additional metrics. Unknown states remain eligible, cannot count as detections, and receive AP score −1, below valid nonnegative scores.

Method	Precision	Recall	AP	Episodes/h	Missed	Known (%)
Persistence	0.203	0.033	0.130	480.8	8	100.0
ARX one-step	0.868	0.748	0.770	1426.3	27	90.0
Following point	0.878	0.747	0.770	793.9	27	90.0
Following hysteresis	0.841	0.735	0.770	99.0	27	90.0
Nominal hysteresis	0.706	0.719	0.587	84.8	27	90.0
Nominal fixed return	0.656	0.658	0.587	48.5	27	90.0
Learned point	0.658	0.748	0.589	1054.5	27	90.0
Learned hysteresis	0.628	0.747	0.589	62.6	27	90.0
Learned fixed return	0.641	0.747	0.589	115.2	27	90.0
Uncertainty return	0.318	0.747	0.589	26.3	27	90.0
Episode CP band	0.742	0.746	0.589	0.0	27	90.0
Matched no allowance	0.675	0.747	0.589	28.3	27	90.0
Matched no dwell	0.325	0.747	0.589	28.3	27	90.0

Table 15: E2 complete normal-reference comparison. Each row has 29,700 eligible observations and 108 forced-latch probes. Selected reference and selected ARX are identical and appear once.

Plant	Reference	RMSE	Allowance	Feasible (%)	Clear/probes
Q1	ARX(4), free	0.00948	0.0553	99.80	108/108
Q1	Selected ARX, free	0.00934	0.0553	99.82	108/108
Q1	Selected FIR	0.01211	0.2322	0.00	0/108
Q1	ARX(4), one-step	0.00976	0.0649	97.92	108/108
Q1	Command as state	0.09584	2.1344	0.00	0/108
Q2	ARX(4), free	0.01020	0.1469	0.00	0/108
Q2	Selected ARX, free	0.00972	0.0716	93.32	108/108
Q2	Selected FIR	0.01693	0.3971	0.00	0/108
Q2	ARX(4), one-step	0.01036	0.1211	0.00	0/108
Q2	Command as state	0.10423	2.1528	0.00	0/108
Q3	ARX(4), free	0.01586	0.1913	0.00	0/108
Q3	Selected ARX, free	0.01439	0.1175	0.00	0/108
Q3	Selected FIR	0.02215	0.4596	0.00	0/108
Q3	ARX(4), one-step	0.01573	0.1807	0.00	0/108
Q3	Command as state	0.11251	2.2058	0.00	0/108

Table 16: Final audit: unadjusted point precision/recall, benign occupancy, alarm starts per benign hour and missed events.

Method	Precision	Recall	Benign (%)	Starts/h	Misses
base point	0.6434	0.9931	0.957	1537.4	0
base return	0.4777	0.9936	7.644	1208.1	0
base long	0.4699	0.9936	8.031	1208.1	0
base dual	0.4699	0.9936	8.031	1208.1	0
linear point	0.6421	0.9930	1.004	1618.2	0
linear dual	0.4651	0.9934	8.415	1270.7	0
RBF point	0.6445	0.9929	0.937	1501.0	0
RBF return	0.4811	0.9932	7.466	1191.9	0
RBF long	0.4729	0.9932	7.870	1191.9	0
RBF dual	0.4729	0.9932	7.870	1191.9	0
RBF dual, no dwell	0.6214	0.9929	1.341	1501.0	0
following point	0.8575	0.9930	0.973	1618.2	0
following dual	0.5487	0.9934	8.569	1331.3	0
RBF following dual	0.5486	0.9934	8.617	1317.2	0
persistence	0.1286	0.0192	0.591	274.7	41
mean (21)	0.0385	0.0046	0.094	78.8	60
base CUSUM	0.2665	0.9887	0.962	301.0	0

The original voraus-AD file is linked by `https://github.com/vorausrobotik/voraus-ad-dataset`. It has 1,115,942,833 bytes and SHA-256 `c90ab1c78af52651b954d41787f7e89d750f0a128b57600b0e5ceec22621f704`. Data are CC BY-NC-SA 4.0; code is MIT. Metadata verify 2,321,690 rows, 137 columns and the 2,122 executions in Table 3. Setting 72 (PRE_A) supplies all 948 training executions. Within-execution time resets to zero, increases strictly with increments 0.00845194–0.01211095 s, and does not establish absolute acquisition order. Maximum train/test lengths are both 1,164 rows. Process signals remain unopened.

Pinned MVT-Flow code `a91a86a642d23df58833b792a53de01edfd81abe` uses $\|z\|^2 - \log|J|$ for training, whereas the paper/scoring use $\frac{1}{2}\|z\|^2 - \log|J|$. The paper describes three convolutions; the released default constructs two. A padding truncation is overwritten, but no test execution in this release exceeds the training length. Test AUROC

Table 17: Primary base-dual versus RBF-dual comparison by plant. Every plant has 29,700 benign points and 36 attack events.

Plant	Base benign	RBF benign	Base gaps	RBF gaps	Base fresh	RBF fresh
Q1	2397	2522	0	0	36	36
Q2	2646	2532	2	3	29	33
Q3	2113	1958	0	0	36	36

is computed each epoch, while the saved checkpoint is the fixed final epoch. These discrepancies remain unresolved; they do not establish test-based selection or invalid published results.

FactoryNet v5 (8 October 2026) remains an unexecuted preprint lead: its inspected newer simulation release is incomplete, imported sources overlap, context contains fault labels and licenses vary. No observations were downloaded; source receipts remain in the audit.

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